

Leveraging Embeddings in Vector Databases



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Vector and Embeddings

Vector

Mathematical object with
magnitude and direction

Very general concept

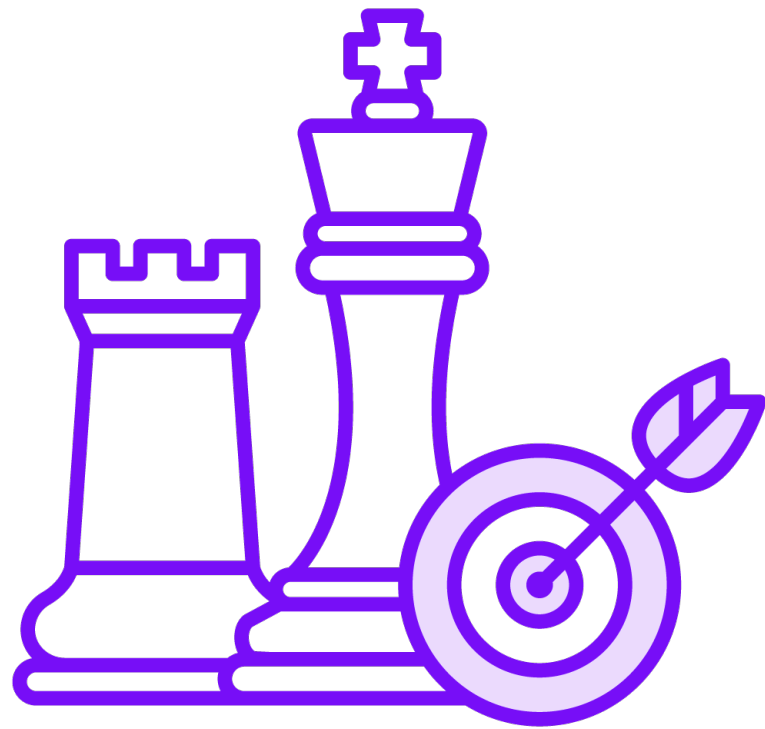
Embeddings

Specific type of vector
representation

Capture complex patterns and
relationships in data



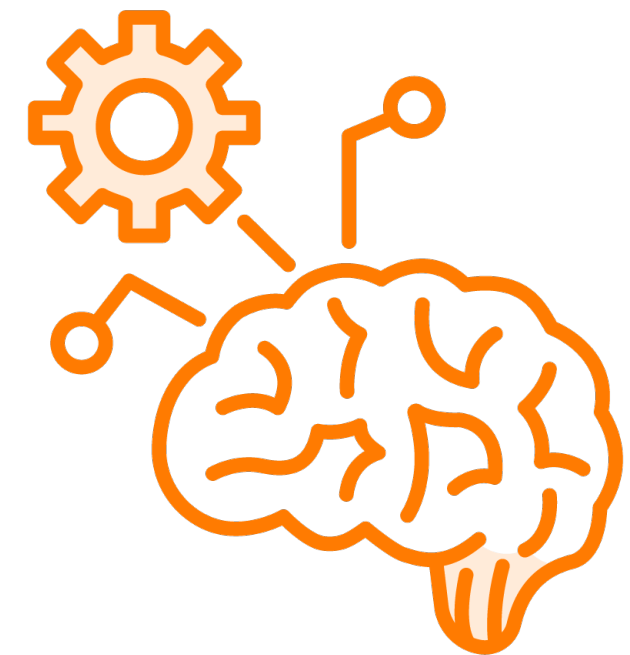
Vector and Embeddings : Key Differences



Scope



Content

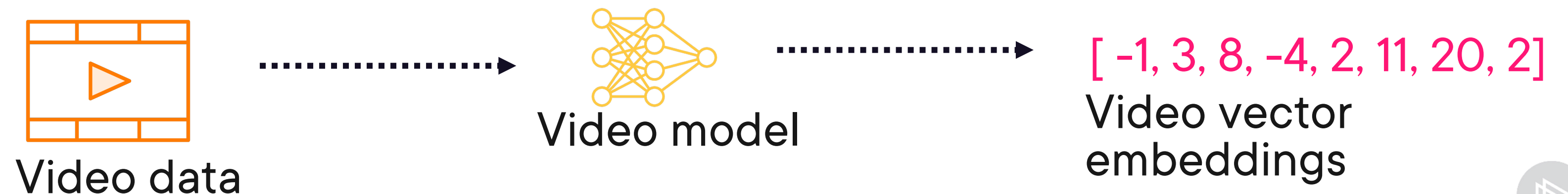
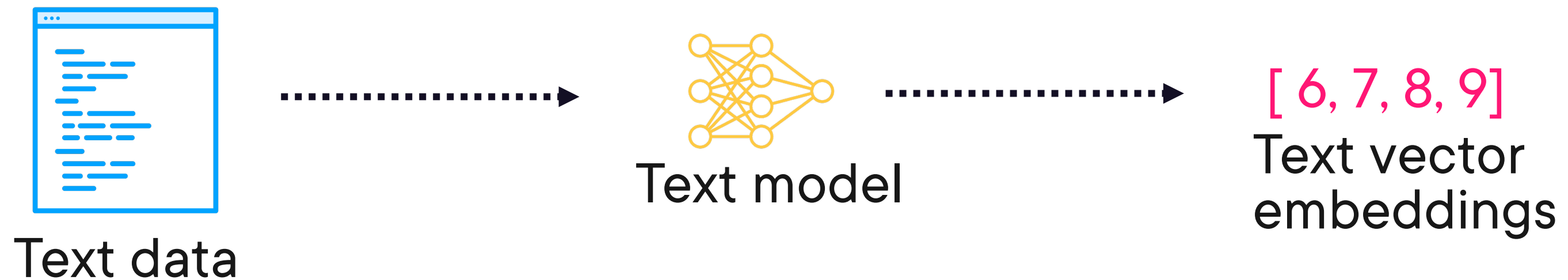


Application

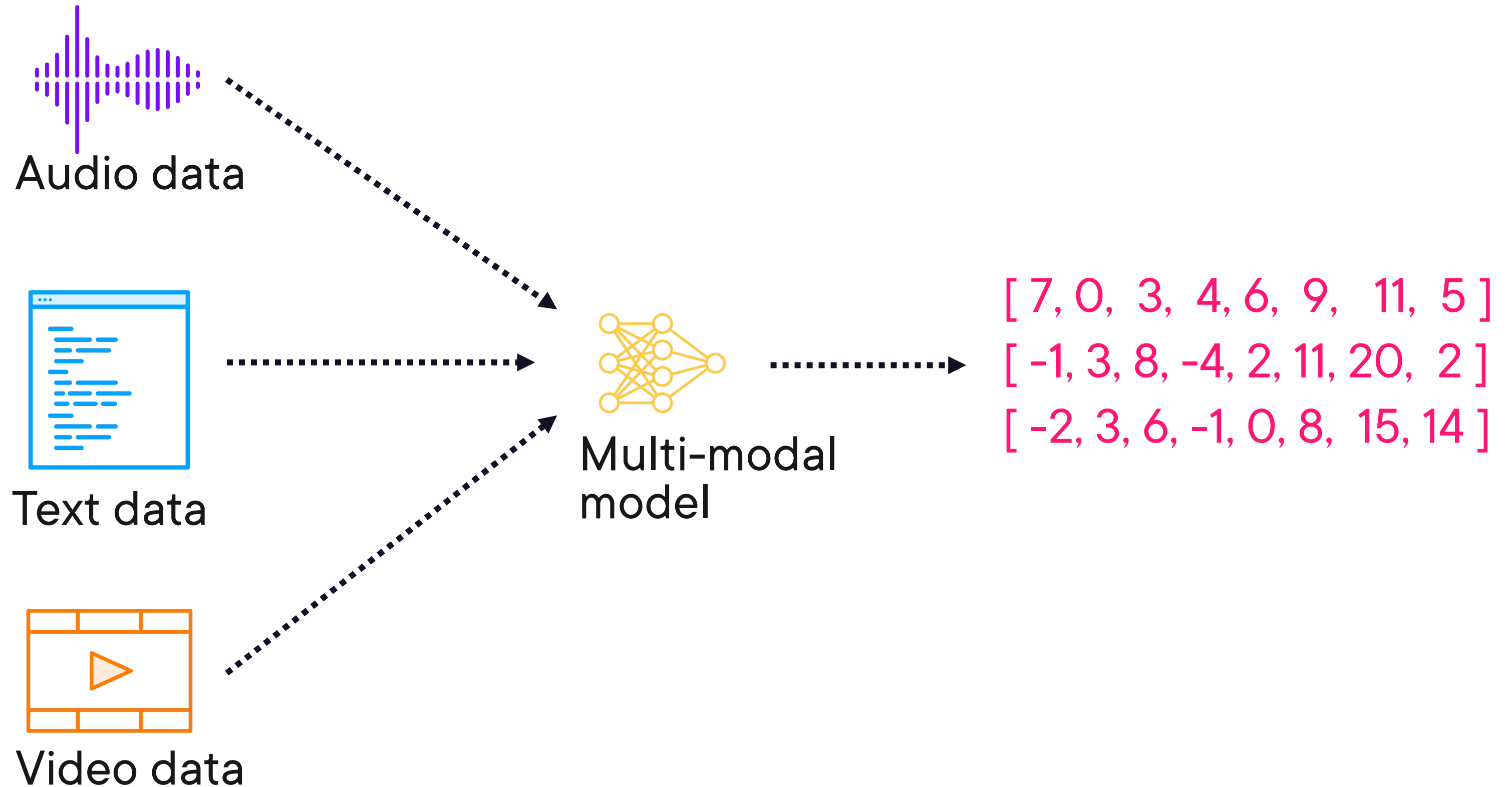
Transforming Data into Embeddings



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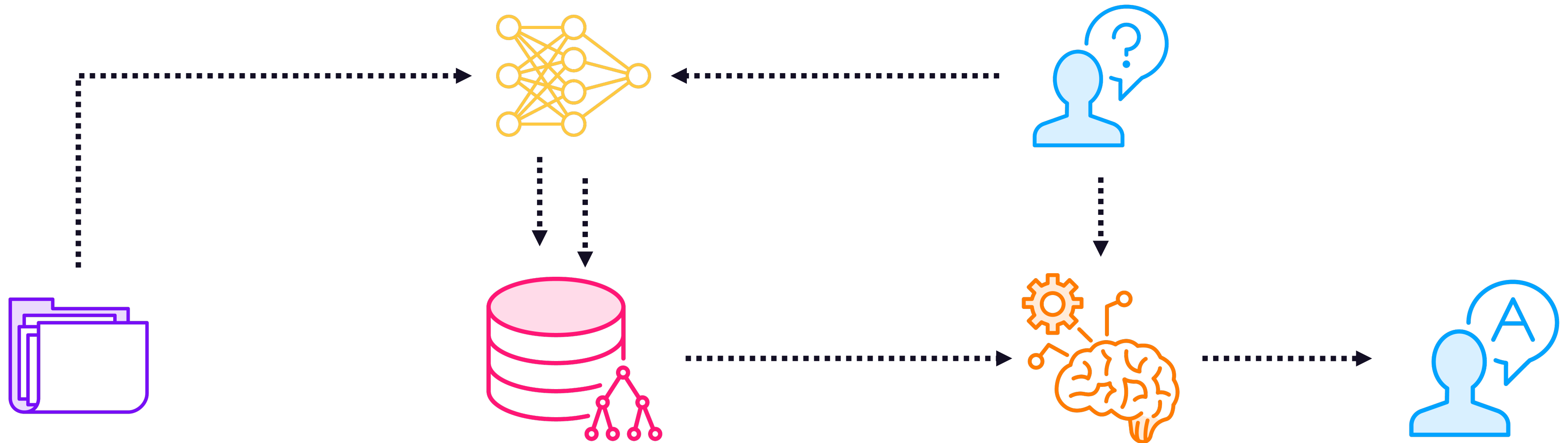




Designing a Vector Application



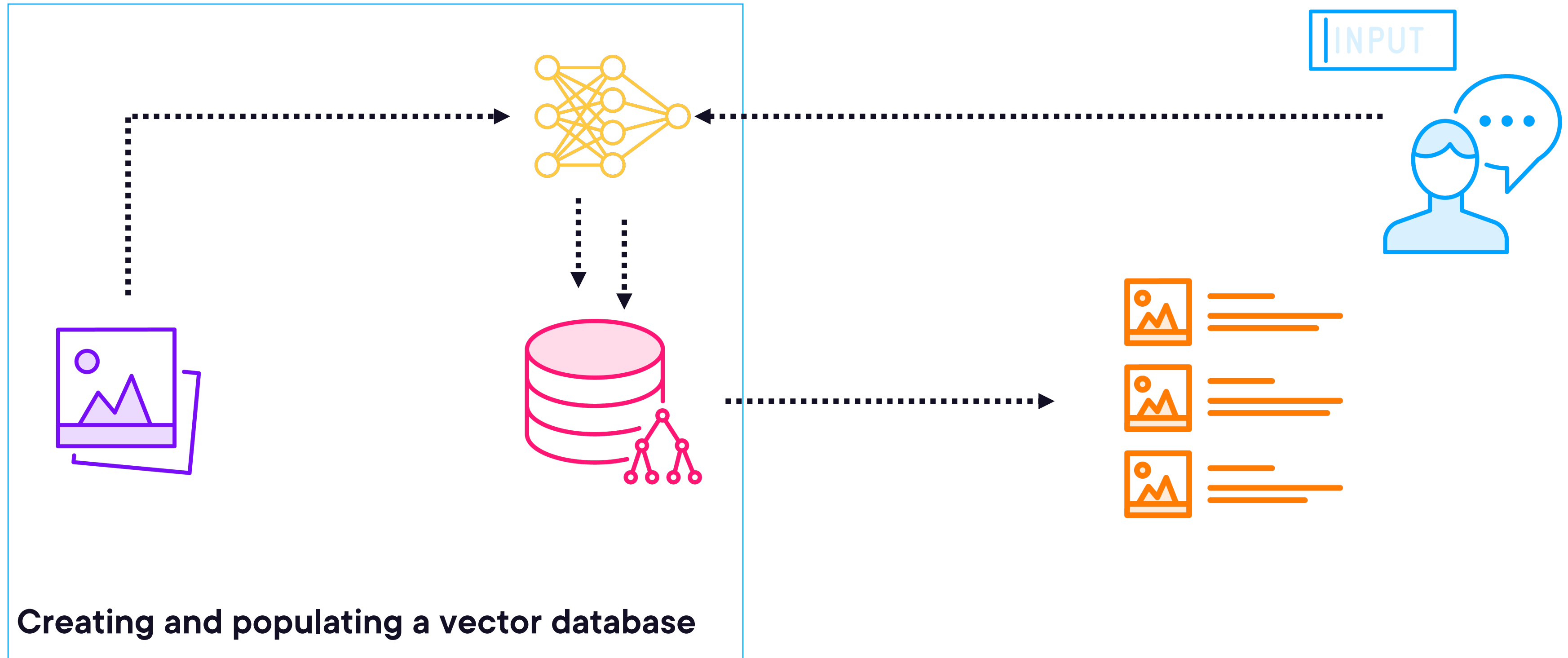
Use Case 1 : Retrieval Augmented Generation (RAG)



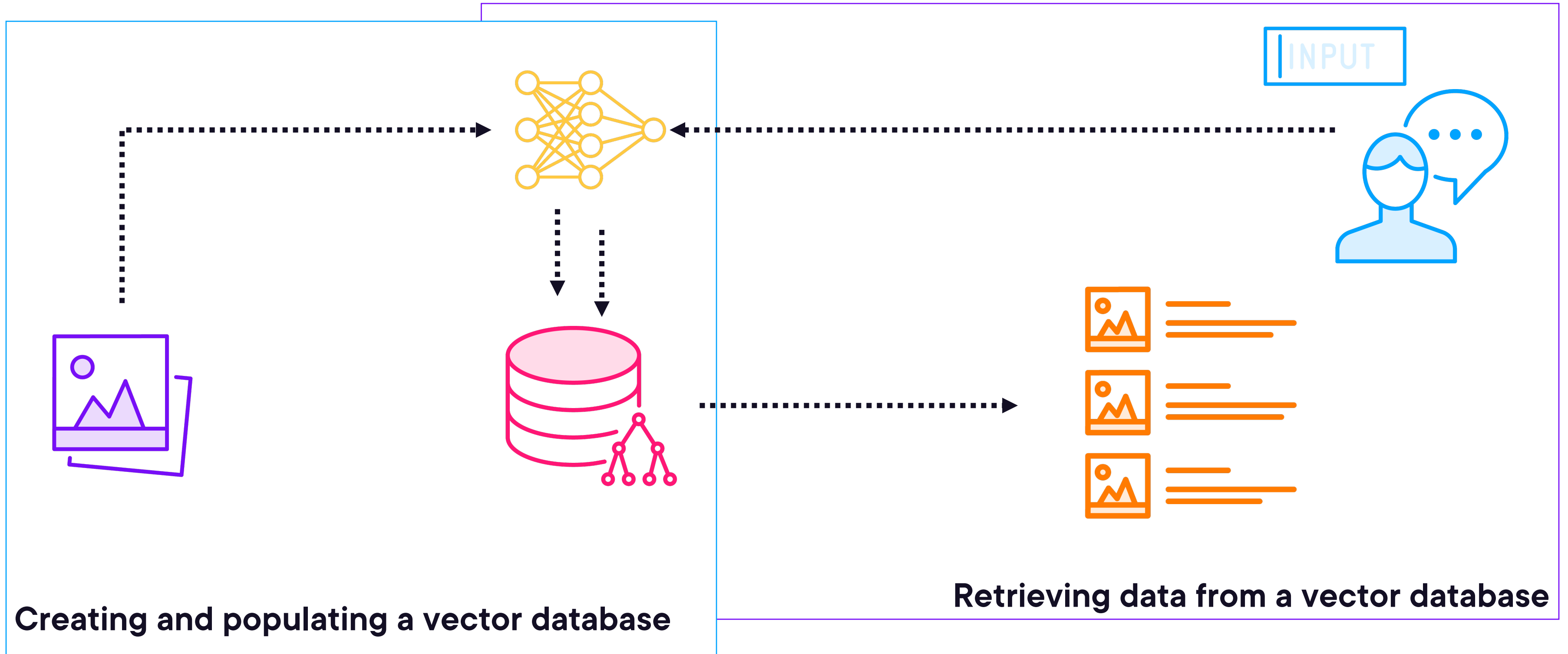
RAGs are not 100% traceable as the generative capability can introduce hallucinations



Use Case 2 : Natural Language Image Retrieval



Use Case 2 : Natural Language Image Retrieval



Demo



Creating and populating a vector database

- Creating index structure
- Populating database with image embeddings



Demo



Retrieving data from a vector database

- Transforming user query into embeddings
- Querying database for the most similar results

